

Standard Operating Procedure **20 Ton Shop Press**

SOP: Redline 20 Ton Air Hydraulic Heavy Duty Shop Press

The Redline 20-Ton Press is a heavy-duty H-frame tool used for installing/removing bearings, bushings, and gears, as well as bending or straightening steel components. It utilizes a dual-operation hydraulic ram that can be powered by either a manual hand pump or an air-actuated foot pedal. This machine exerts up to **40,000 lbs** of force; failure to center loads or exceed capacity can result in catastrophic structural failure or projectile ejection.

Potential Hazards

- **Mechanical (High Pressure):** Stored energy can cause parts to "pop" or fly out of the press with deadly force if the load is not perfectly centered.
- **Crush Points:** Extreme pinch points exist between the ram and the workpiece, and between the workpiece and the bed plates.
- **Structural Failure:** Overloading the frame or using improper pins/mandrels can cause the press to deform or collapse.
- **Pneumatic/Hydraulic:** High-pressure air lines and hydraulic fluid leaks (pinhole leaks can penetrate skin).

Personal Protective Equipment (PPE)

PPE Item	Requirement
Eye Protection	**Safety Goggles (ANSI Z87.1) + Face Shield:** Mandatory. Under high pressure, brittle metals or bearings can shatter and eject at ballistic velocities.
Foot Protection	**Steel-Toed Boots:** Mandatory. Heavy arbor plates (V-blocks) and workpieces present significant crush hazards if dropped.
Hand Protection	**Heavy-Duty Work Gloves:** Recommended for handling heavy steel stock, though fingers must stay clear of the ram's travel path.
Apparel	Secure loose clothing and lanyards. Remove all jewelry, particularly rings and watches, which can be snagged during setup.



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Before Use

- **Capacity Check:** Confirm the task does not exceed the **20-ton** limit. Never use cheater bars on the manual handle to increase leverage.
- **Hardware Inspection:** Ensure the **bed support pins** are fully inserted and locked through both sides of the H-frame.
- **Ram & Hose Check:** Inspect the hydraulic ram for leaks and the air hose for frays or bulges.
- **Bed Alignment:** Ensure the press bed is level and the arbor plates (V-blocks) are positioned securely and in a way that supports the workpiece without slipping.
- **Mandrel Selection:** Use only high-strength steel mandrels or "pushers." Never use hardened chrome sockets, as they can shatter under pressure.

During Use

- **The "Center Rule":** Always center the workpiece directly under the ram. Offset loading will cause the ram to "kick" or bend, potentially ejecting the part.
- **Initial Contact:** Lower the ram slowly using the manual handle until it makes contact with the workpiece. Double-check alignment before applying significant pressure.
- **Air Operation:** Use the air foot pedal for consistent pressure, but keep your hands completely clear of the work area.
- **Gauge Monitoring:** Watch the integrated pressure gauge. If the tonnage stops rising but the ram continues to move, the workpiece or the press frame is deforming. **Stop immediately.**
- **The "Shield" Rule:** Whenever possible, place a heavy-duty Lexan shield or a reinforced mesh screen in front of the press work area during high-pressure operations.
- **Sudden Release:** Be prepared for a loud "bang" when a bearing or bushing breaks free; this is normal, but maintain your stance and do not flinch.

After Use

- **Ram Retraction:** Slowly open the release valve to retract the ram. Do not allow the ram to "snap" back up.



- **Cleanup:** Remove arbor plates and return them to their storage location. Clear any metal shavings or leaked hydraulic fluid from the floor.
- **Housekeeping:** Sweep the area around the press base. Ensure the air supply is disconnected if the machine will be idle for an extended period.
- **Material Management:** Deburr any parts that were deformed during the pressing process.

Emergency Equipment and Procedures

- **Hydraulic Failure:** If a hose bursts, release the air pedal and stay back until pressure has dissipated.
- **Projectile Incident:** If a part is ejected, do not attempt to catch it. Clear the area and report the incident to the Laboratory Project Supervisor.
- **Reporting:** Report all "near-misses," bent bed pins, or hydraulic leaks immediately.